Type MS · High-level and low-level languages

Answer 1 of 5 · Question 5 part b.ii · 2 marks

2025 JUN · Paper 01 · extract 1 of 5 · find part b.ii in this mark scheme extract

5(a)(i)	Award **one** mark for any of the following up to a maximum of **two** marks: • Heart rate (1) to measure blood flow/pulse rate (1) • Barometric pressure (1) to determine altitude (1) • Microphone (1) for voice activation control (1) • Temperature/Thermometer (1) to measure ambient/body temperature (1) • GPS (1) to calculate longitude/latitude (1) • Light (1) to adjust screen brightness/charge watch battery (1) • Magnetometer (1) to act as a compass (1) • ECG (1) measure heart rhythm (1) • Blood pressure monitor (1) measure pressure in arteries / can report heart rate/pulse rate (1)	Allow any reasonable sensor that might be embedded within a smart watch. Purpose must be linked to the sensor Purpose must expand on name (e.g. heart rate sensor - monitor user's heart rate is not sufficient for second mark) Allow purpose mentioned in second column if nothing in first column	**2**
Question	Answer	Additional Guidance	Mark
5(a)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: • It has a low power requirement (1) • It can use a low processor speed (1) • It has built in/integrated storage (1) • It relies on flash memory (1) • It can be mass produced at a low cost per device (1) • It is small (1) • It can work in a large range of operating temperature ranges (1) • It is robust (1) • It has its operating software set by the manufacturer (1) • It is designed to do one particular task (1)		**2**

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Type MS · High-level and low-level languages

Answer 1 of 5 · Question 5 part b.ii · 2 marks

2025 JUN · Paper 01 · extract 2 of 5 · find part b.ii in this mark scheme extract

Question	Answer	Additional Guidance	Mark
5(b)(i)	**The only correct answer is B** *A is not correct because low-level languages are not converted into high-level languages* *C is not correct because assemblers translate assembly code to produce machine code but do not execute it* *D is not correct because compilers and interpreters translate high-level languages*		1
Question	Answer	Additional Guidance	Mark
5(b)(ii)	Award up to **two** marks for a linked explanation such as: • Makes best use of the hardware (1) because they can directly access (specific) hardware (features) (1) • It reduces manufacturing costs (1) because of a lower processor specification/expense (1) • Optimise their code for hardware (1) directly access/control registers (1)	Do not accept references to translation time	2
Question	Answer	Additional Guidance	Mark
5(b)(iii)	Award **one** mark for any of the following up to a maximum of **two** marks: • Abide by the law / avoid plagiarism / copyright infringements (1) • Adhere to a professional code of conduct (1) • Maintain/enhance their skills through further learning/keep up to date/attend (relevant) conferences (1) • Test code thoroughly/do not release code with known errors/vulnerabilities (1) • Respect privacy and security of data (1) • Preserve the environment (1) • Make software accessible/unbiased (1)		2

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Type MS · High-level and low-level languages

Answer 1 of 5 · Question 5 part b.ii · 2 marks

2025 JUN · Paper 01 · extract 3 of 5 · find part b.ii in this mark scheme extract

Question	Answer	Additional Guidance	Mark
5(b)(iv)	**The only correct answer is A** *B is not correct because 500 KiB is 512,000 bytes* *C is not correct because 0.5 MiB is 524,288 bytes* *D is not correct because 5MB is 5,000,000 bytes*		**1**
Question	**Answer**	**Additional Guidance**	**Mark**
5(c)	Indicative content: • Can use sensors to take/track readings such as heart rate • Can track location via GPS and record routes taken/distance travelled • Can calculate exercise/movement undertaken such as steps/calories burned • Can be linked via 5G/Internet Of Things/Personal Area Network (PAN) to other devices/apps that can record and analyse the data • Can trigger alerts when users are inactive • Can monitor blood oxygen/blood sugar/sleep patterns • Can applaud/give kudos when an achievement/event takes place • Can increase motivation of the wearer to be healthy/perform exercise due to feedback loop • Workouts/alerts can be synced to the watch • Linked apps can perform more in-depth analysis • No guarantee that users will buy into benefits and lead a healthier lifestyle • Monitoring vulnerable people allowing independent living	Allow content referring to other wearable technology which is not a smartwatch	**6**
		Total for question 5	**16**

Type MS · High-level and low-level languages

Answer 1 of 5 · Question 5 part b.ii · 2 marks

2025 JUN · Paper 01 · extract 4 of 5 · find part b.ii in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1-2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.

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Type MS · High-level and low-level languages

Answer 1 of 5 · Question 5 part b.ii · 2 marks

2025 JUN · Paper 01 · extract 5 of 5 · find part b.ii in this mark scheme extract

Question	Answer	Additional Guidance	Mark

Type MS · High-level and low-level languages

Answer 2 of 5 · Question 6 part b · 4 marks

2019 JUN · Paper 01 · extract 1 of 5 · find part b in this mark scheme extract

6(a)(i)	Features such as power consumption / processor / memory / number and type of components / physical size (1) can be optimised / chosen to fit just one specific task. (1)		2

Question Number	Answer	Additional Guidance	Mark
6(a)(ii)	Any **two** from: • Printer (1) • Bank card (magnetic strip) reader/scanner (1) • Bank card contactless/NFC reader/scanner (1) • Cash input and giving change device (1)	Allow any hardware that relates to paying by cash for MP4	2

Question Number	Answer	Additional Guidance	Mark
6(a)(iii)	To prevent unauthorised people from reading / understanding the bank card information (1)	Do not award 'to keep it from being hacked / accessed'.	1

Type MS · High-level and low-level languages

Answer 2 of 5 · Question 6 part b · 4 marks

2019 JUN · Paper 01 · extract 2 of 5 · find part b in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)	Award **one** mark for each of (maximum of 4): 1. Readability: high-level language is easily readable by a human because it is designed to look like English whereas a low-level language is binary patterns / mnemonics / abbreviations. (1) 2. Portability: code written in a high-level language is easily transportable across machines whereas a low-level language is designed for a specific microprocessor / piece of hardware. (1) 3. Uses: low-level languages are commonly used to write programs for hardware or devices, like printers, whereas high-level languages are commonly used for to write higher-level applications, such as word processors. (1) 4. Optimisation: low-level languages are used to improve performance, especially of hardware, whereas high-level languages have to go through multiple layers of software to get to the hardware, e.g. the need for HLL to be compiled/interpreted makes programmes run slower than LLL, HLL has a higher level of abstraction removing unnecessary detail (one to many), LLL keeps more of a detail (one to one) (1) 5. Ease of use: high-level languages usually have tools, functions, libraries and development environments which are accessible, even by those with little knowledge, whereas, low-level languages are often challenging to work with and require expertise and depth of knowledge. (1)	For each mark point: • the general feature is followed by exemplification • needs to be a comparison, which may be implicit	4

Type MS · High-level and low-level languages

Answer 2 of 5 · Question 6 part b · 4 marks

2019 JUN · Paper 01 · extract 3 of 5 · find part b in this mark scheme extract

Question Number	Answer	Mark
6(c)	Indicative content: Characteristics – Any of the following: • Implemented in software. • Could involve the use of neural networks / clustering / modelling • Algorithms are trained by exposure to data that has been validated / checked • Algorithms learn by identifying commonalities between the validated / checked data • Algorithms make predictions for new data based on prior learning • Algorithms may incorporate recursive self-improvement Uses – Any of the following: • Game playing (opponents) • Analytics (analyse buying patterns, predicting behaviours, predictive text, targeted advertising, personalised news feeds, friend suggestions) • Image / Audio processing (recognising objects/patterns, face recognition) • Logistics (scheduling, order fulfilment) • Control systems (cars, manufacturing, weapons, navigation • Diagnostic systems (medical, mechanical, electrical) • Natural languages processing (chatterbots, chatbots, speech recognition, personal assistants) • Robotics (dangerous situations, help aged or disabled) Ethical issues – Any of the following: • Training sets could include data that should not be used to make decisions (race, religion, gender, etc.) that could cause biased outcomes • Is a computer to be trusted to make decisions (life-death)? • If a computer discovers something that humans can't prove, should it be accepted as truth? • How do we know that the answer produced is correct? • Will people be comfortable interacting with machines that are considered intelligent? • Some jobs may be at risk, especially those requiring human skills such as empathy or creativity.	6

Type MS · High-level and low-level languages

Answer 2 of 5 · Question 6 part b · 4 marks

2019 JUN · Paper 01 · extract 4 of 5 · find part b in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent and logically structured.

Type MS · High-level and low-level languages

Answer 2 of 5 · Question 6 part b · 4 marks

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Type MS · High-level and low-level languages

Answer 3 of 5 · Question 6 part a.i · 1 mark

2021 NOV · Paper 01 · extract 1 of 6 · find part a.i in this mark scheme extract

6(a)(i)	Award **one** mark from: • faster translation/execution (1) • allows her to directly address components / make efficient use of available memory space (1) • can dispense with the need for an OS to make the code run faster/free up memory space (1)		1
6(a)(ii)	Award **one** mark from: • difficult to read/understand (1) • easy to make mistakes (1) • can be hard to find errors in the code (1) • time consuming (to write) (1) • not very portable / processor-specific (1) • lack of built-in functions/procedures (1) • few development/editorial tools available (1)		1

Type MS · High-level and low-level languages

Answer 3 of 5 · Question 6 part a.i · 1 mark

2021 NOV · Paper 01 · extract 2 of 6 · find part a.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)(i)	Award **one** mark from: • virus detection/scanner (1) • threat identification (1) • real-time scanning (1) • scan scheduling (1) • quarantine files (1) • carry out automatic updates (1) • on-demand file scanning (1) • kill switch (1)		2
6(b)(ii)	Award **one** mark from: • anti-spyware (1) • anti-adware (1)		1
6(c)	A AND O AND (W OR D) Award **one** mark each to a maximum of **four** marks for: • A AND O (1) • W OR D (1) • Brackets around W OR D (1) • AND between A AND O (W OR D) (1)		4

Type MS · High-level and low-level languages

Answer 3 of 5 · Question 6 part a.i · 1 mark

2021 NOV · Paper 01 · extract 3 of 6 · find part a.i in this mark scheme extract

6(d)	**Intellectual property** • Intellectual property is any work that is distinct, owned, and protected by patent or copyright laws • Akiko's software programs fall into this category • She will be able to protect her intellectual property • Her software is protected by copyright without her having to apply for it **Licensing** • Purchasing software does not mean you own it • Could add a licence key to the software. • Could require compulsory registration using the internet • Could allow proprietary licences, which do not allow code modification or code reuse. • Could allow Free and open-source software (FOSS), which would allow the user to modify and reuse the code. • Creative Commons (CC) ○ Could use creative commons (CC) licences that would allow the free distribution of copyrighted work ○ Would use if Akiko wanted to allow people to share, use and build on the programs. ○ May mention some of the CC licences and conditions	6

Type MS · High-level and low-level languages

Answer 3 of 5 · Question 6 part a.i · 1 mark

2021 NOV · Paper 01 · extract 4 of 6 · find part a.i in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
Level 2	3-4	Demonstrates adequate knowledge and understanding of key concepts/principles of computer science. The discussion shows some linkages and lines of reasoning with some structure.
Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.
		Total **15**

Type MS · High-level and low-level languages

Answer 3 of 5 · Question 6 part a.i · 1 mark

2021 NOV · Paper 01 · extract 5 of 6 · find part a.i in this mark scheme extract

Type MS · High-level and low-level languages

Answer 3 of 5 · Question 6 part a.i · 1 mark

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Type MS · High-level and low-level languages

Answer 4 of 5 · Question 6 part a.ii · 1 mark

2021 NOV · Paper 01 · extract 1 of 6 · find part a.ii in this mark scheme extract

6(a)(i)	Award **one** mark from: • faster translation/execution (1) • allows her to directly address components / make efficient use of available memory space (1) • can dispense with the need for an OS to make the code run faster/free up memory space (1)		1
6(a)(ii)	Award **one** mark from: • difficult to read/understand (1) • easy to make mistakes (1) • can be hard to find errors in the code (1) • time consuming (to write) (1) • not very portable / processor-specific (1) • lack of built-in functions/procedures (1) • few development/editorial tools available (1)		1

Type MS · High-level and low-level languages

Answer 4 of 5 · Question 6 part a.ii · 1 mark

2021 NOV · Paper 01 · extract 2 of 6 · find part a.ii in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
6(b)(i)	Award **one** mark from: • virus detection/scanner (1) • threat identification (1) • real-time scanning (1) • scan scheduling (1) • quarantine files (1) • carry out automatic updates (1) • on-demand file scanning (1) • kill switch (1)		2
6(b)(ii)	Award **one** mark from: • anti-spyware (1) • anti-adware (1)		1
6(c)	A AND O AND (W OR D) Award **one** mark each to a maximum of **four** marks for: • A AND O (1) • W OR D (1) • Brackets around W OR D (1) • AND between A AND O (W OR D) (1)		4

Type MS · High-level and low-level languages

Answer 4 of 5 · Question 6 part a.ii · 1 mark

2021 NOV · Paper 01 · extract 3 of 6 · find part a.ii in this mark scheme extract

6(d)	**Intellectual property** • Intellectual property is any work that is distinct, owned, and protected by patent or copyright laws • Akiko's software programs fall into this category • She will be able to protect her intellectual property • Her software is protected by copyright without her having to apply for it **Licensing** • Purchasing software does not mean you own it • Could add a licence key to the software. • Could require compulsory registration using the internet • Could allow proprietary licences, which do not allow code modification or code reuse. • Could allow Free and open-source software (FOSS), which would allow the user to modify and reuse the code. • Creative Commons (CC) ○ Could use creative commons (CC) licences that would allow the free distribution of copyrighted work ○ Would use if Akiko wanted to allow people to share, use and build on the programs. ○ May mention some of the CC licences and conditions	6

Type MS · High-level and low-level languages

Answer 4 of 5 · Question 6 part a.ii · 1 mark

2021 NOV · Paper 01 · extract 4 of 6 · find part a.ii in this mark scheme extract

Level	Mark	Descriptor
	0	No rewardable content.
Level 1	1–2	Basic, independent points are made showing elements of knowledge and understanding of key concepts/principles of computer science. The discussion will contain basic information with little linkage between points made.
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Level 3	5-6	Demonstrates comprehensive knowledge and understanding by selecting relevant knowledge and understanding of key concepts/principles of computer science to support the discussion being presented. The discussion shows a well-developed, sustained line of reasoning which is clear, coherent, and logically structured.
		Total **15**

Type MS · High-level and low-level languages

Answer 4 of 5 · Question 6 part a.ii · 1 mark

2021 NOV · Paper 01 · extract 5 of 6 · find part a.ii in this mark scheme extract

Type MS · High-level and low-level languages

Answer 4 of 5 · Question 6 part a.ii · 1 mark

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Type MS · High-level and low-level languages

Answer 5 of 5 · Question 1 part e.i · 1 mark

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1(a)(i)	Award **one** mark from: • An action/task that is carried out (by the CPU) (1) • An operation/code that is/will be executed (by the CPU) (1) • Something that tells the CPU/processor/device what to do (1)	Allow examples	1
1 (a)(ii)	Award **one** mark from: • A pointer/reference to/identification of a location in memory (1) • A pointer/reference to/location where data will be accessed from/stored (to)(1)		1

Question Number	Answer	Additional Guidance	Mark
1(b)	**The only correct answer is D** *A is not correct because the address bus carries the address of the memory location* *B is not correct because the data bus carries but does not store data* *C is not correct because the control unit sends signals but does not handle the data*		1

Question Number	Answer	Additional Guidance	Mark
1(c)(i)	Award **one** mark from: • More instructions can be carried out per second (1) • Processes run faster (1) • Programs executed faster (1) • Can run more complex programs (1)		1
1(c)(ii)	Award **one** mark from: • The CPU/computer could overheat (1) • More cooling required (1) • Needs more power (1) • CPU could become unstable/crash / its lifespan could be shortened (1)		1

Type MS · High-level and low-level languages

Answer 5 of 5 · Question 1 part e.i · 1 mark

2022 JUN · Paper 01 · extract 2 of 4 · find part e.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark
1(d)	**The only correct answer is D** *A is not correct because it describes a method of accessing data from a secondary storage device* *B is not correct because it describes the multi-agent computational model* *C is not correct because it describes the parallel computational model*		1

Type MS · High-level and low-level languages

Answer 5 of 5 · Question 1 part e.i · 1 mark

2022 JUN · Paper 01 · extract 3 of 4 · find part e.i in this mark scheme extract

<table>
<tr><th>Question Number</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>1(e)(i)</td><td>Award one mark from:

• When fast execution of the program is essential/gives faster execution (1)
• When writing code to control a device / directly access the hardware (1)
• To make efficient use of a device's limited storage/memory/power (1)
• When writing code for a particular architecture/hardware/embedded system (1)</td><td></td><td>1</td></tr>
<tr><td>1(e)(ii)</td><td>Award one mark for:

• To translate/convert assembly language into machine/object code (1)</td><td></td><td>1</td></tr>
<tr><td>1(e)(iii)</td><td>Award one mark for any two correct rows (1)
Award two marks for any four correct rows (2)
Award three marks for all five correct rows (3)

<table><tr><th>Description</th><th>Compiler</th><th>Interpreter</th></tr><tr><td>Translates the program each time it is executed</td><td></td><td>☐</td></tr><tr><td>Produces permanent object code</td><td>☐</td><td></td></tr><tr><td>Translates line by line</td><td></td><td>☐</td></tr><tr><td>Translates the whole program before it is run</td><td>☐</td><td></td></tr><tr><td>Generates a list of errors once the complete program has been translated</td><td>☐</td><td></td></tr></table></td><td>Row can only be correct if there is only one tick.</td><td>3</td></tr>
<tr><td colspan="3">Total for question 1</td><td>11</td></tr>
</table>

Type MS · High-level and low-level languages

Answer 5 of 5 · Question 1 part e.i · 1 mark

2022 JUN · Paper 01 · extract 4 of 4 · find part e.i in this mark scheme extract

Question Number	Answer	Additional Guidance	Mark